

1 Post-Stratification Weights

1.1 Intro

It requires the use of auxiliary information about the population and may take a number of different variables into account. Information usually needed:

- Population estimates of the distribution of a set of demographic characteristics that have also been measured in the sample
- For example, information found in the Census (as a measure of population) such as:
 - Sex
 - Age
 - Education attainment
 - Household size
 - Residence
 - Region
- Population data for community-based samples:
 - US Census tabulations
 - The Current Population Survey (CPS)
 - The American Community Survey (ACS)

1.2 Single-variable dataset



Calculating Post-Stratification Weights

Gender	Population Proportion	Sample Proportion	Population / Sample	Weight
Female	.5	.6	.5 / .6	.8333
Male	.5	.4	.5 / .4	1.25
Total	1	1		

Census report is used to find the gender distribution in the population (50% female). This is compared to the gender distribution in the sample of completed interviews (60% female).

Problem: *What if you have more than one characteristic to balance with the population?*

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1.3 Multi-variable dataset



Adjusting for Multiple Population Characteristics

- **Options for combining characteristics:**
 - You can combine characteristics in a single table to do the calculation:
 - Males 18-25
 - Males 26-45
 - Males 46+
 - Females 18-25
 - Females 26-45
 - Females 46+
- **However:**
 - You need to have these crosstab tables available for the population source
 - The number of cases in each cell in the sample cannot be too small.
- **Therefore:** It may be better to use several separate frequency tables rather than one big N-way crosstab to compute the weights, especially when several characteristics are being balanced.

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How to compute the weight given multiple characteristics features?



Three Possible Solutions

- 1. Use a single big age x gender X education table for the calculation of the weights.
 - However, crosstabs may not be available for the population
 - and, small cell sizes in the sample table
- 2. Iterative Solutions:
 - Manual version (stepwise programming in statistical software)
 - Automatic version (i.e. Raking software)
- 3. Logistic regression based solutions if case level population data is available.

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1.4 Method 1: Crosstabs

Suppose the population and survey data contain two characteristics features, sex and education.

- Sex: male (sex = 1), female (sex = 0)
- edu: high edu (edu = 1), low edu (edu = 0)

Here is an example of the population and survey data

population			Survey		
	male	edu		male	edu
0	1	1	0	0	1
1	0	1	1	0	1
2	1	1	2	0	1
3	0	1	3	0	1
4	1	1	4	1	1
5	0	0	5	1	1
6	1	0	6	1	0
7	0	0	7	1	0
8	1	0	8	1	0
9	0	0	9	1	0

Hence, we can group obs into four groups

- female with low edu (code: 00)
- female with high edu (code: 01)
- male with low edu (code: 10)
- male with high edu (code: 11)

Here is the percentage share of each group,

$$share_{\text{group}} = \frac{N_{\text{group}}}{N_{\text{num. of obs}}}.$$

For example, there are 2 female with high edu, and there are 10 obs. Hence $share_{\text{female with high edu}} = \frac{2}{10} = 0.2$

Here is the percentage share for each group in the population and the survey.

	population	survey
groups_00	0.3	NaN
groups_01	0.2	0.4
groups_10	0.2	0.4
groups_11	0.3	0.2

NaN says there is no female with low edu (code: 00) in the survey.

To make the demographic feature in the survey being consistent with the population, the weight for group g is defined as

$$weight_g = \frac{\text{share in population}_g}{\text{share in survey}_g}. \quad (1)$$

For example, the weight for female with high edu (code: 01) is

$$\begin{aligned}
 weight_{\text{female with high edu}} &= \frac{\text{share in population}_{\text{female with high edu}}}{\text{share in survey}_{\text{female with high edu}}} \\
 &= \frac{0.2}{0.4} \\
 &= 0.5
 \end{aligned}$$

Here is the crosstab weights for each group.

	population	survey	WT_crosstabs
groups_00	0.3	NaN	NaN
groups_01	0.2	0.4	0.5
groups_10	0.2	0.4	0.5
groups_11	0.3	0.2	1.5

1.5 Method 3: Logistic regression

Given the population and survey data in “Method 1”,

population			Survey		
male	edu		male	edu	
0	1	1	0	0	1
1	0	1	1	0	1
2	1	1	2	0	1
3	0	1	3	0	1
4	1	1	4	1	1
5	0	0	5	1	1
6	1	0	6	1	0
7	0	0	7	1	0
8	1	0	8	1	0
9	0	0	9	1	0

Add a column called ”sample” set to 1 for the survey data, and set to 0 for the population data.

population				survey			
male	edu	sample		male	edu	sample	
0	1	1	0	0	0	1	1
1	0	1	0	1	0	1	1
2	1	1	0	2	0	1	1
3	0	1	0	3	0	1	1
4	1	1	0	4	1	1	1
5	0	0	0	5	1	1	1
6	1	0	0	6	1	0	1
7	0	0	0	7	1	0	1
8	1	0	0	8	1	0	1
9	0	0	0	9	1	0	1

Then merge the two dataset vertically. And fit the equation below,

$$sample = \beta_0 + \beta_1 male + \beta_2 edu. \quad (2)$$

Then get the estimated probability

$$Prob(Sample = 1|\mathbf{x}) = \frac{\exp(\mathbf{x}'\boldsymbol{\beta})}{1 + \exp(\mathbf{x}'\boldsymbol{\beta})} = \Lambda(\mathbf{x}'\boldsymbol{\beta}) = p,$$

by calling “smf.logit().fit().predict()”.

The odds is

$$Odds = \frac{Prob(Sample = 1|\mathbf{x})}{Prob(Sample = 0|\mathbf{x})} = \frac{p}{1 - p} = \exp(\mathbf{x}'\boldsymbol{\beta})$$

The weight for individual i is the inverse of the odds

$$weight_i = \frac{1 - p}{p} \quad (3)$$

	male	edu	groups	WT_crosstabs	WT_logit
0	0	1	01	0.5	1.054032
1	0	1	01	0.5	1.054032
2	0	1	01	0.5	1.054032
3	0	1	01	0.5	1.054032
4	1	1	11	1.5	0.623948
5	1	1	11	1.5	0.623948
6	1	0	10	0.5	1.054032
7	1	0	10	0.5	1.054032
8	1	0	10	0.5	1.054032
9	1	0	10	0.5	1.054032

You may find that the crosstabs weight and the logit weight are different!

Here is why:

- **Crosstabs Weighting (Fully Saturated):** This method treats every combination (e.g., Male + Edu) as a unique, independent island. It accounts for all interactions automatically. It doesn't care if "PhD" behaves similarly for Males and Females; it calculates them separately.
- **Logistic Regression (Additive):** In a standard model ($\text{sample} \sim \text{male} + \text{edu}$), the model assumes these variables are independent. It calculates a "bonus" for being Male and a "bonus" for having a PhD, then adds them together.
- **The Difference:** If PhD holders are over-represented specifically among women but not among men, Crosstabs Weighting will catch that nuance. Logistic Regression will "average" the PhD effect across both genders unless you explicitly add an interaction term (e.g., $\text{male} * \text{edu}$).

Now include the interaction terms:

$$\text{sample} = \beta_0 + \beta_1 \text{male} + \beta_2 \text{edu} + \beta_3 \text{male} \times \text{edu}. \quad (4)$$

In python, you can set the equation as $\text{sample} \sim \text{male} * \text{edu}$. It will include both the main and the interaction effects.

Now the two weights are the same.

	Survey dataset				
	male	edu	groups	WT_crosstabs	WT_logit
0	0	1	01	0.5	0.5
1	0	1	01	0.5	0.5
2	0	1	01	0.5	0.5
3	0	1	01	0.5	0.5
4	1	1	11	1.5	1.5
5	1	1	11	1.5	1.5
6	1	0	10	0.5	0.5
7	1	0	10	0.5	0.5
8	1	0	10	0.5	0.5
9	1	0	10	0.5	0.5

1.6 Advantage of logistic regression

- It can include not only interaction terms, but multinomial terms, e.g., age^2 .
- It can handle continuous variable without creating intervals, such as age bins (21-25, 26-30).
- Crosstabs groups may not be available for the population, and small cell sizes in the sample table.